A data-driven parameterization of potential energy conversion rate of vertical mixing due to Langmuir turbulence

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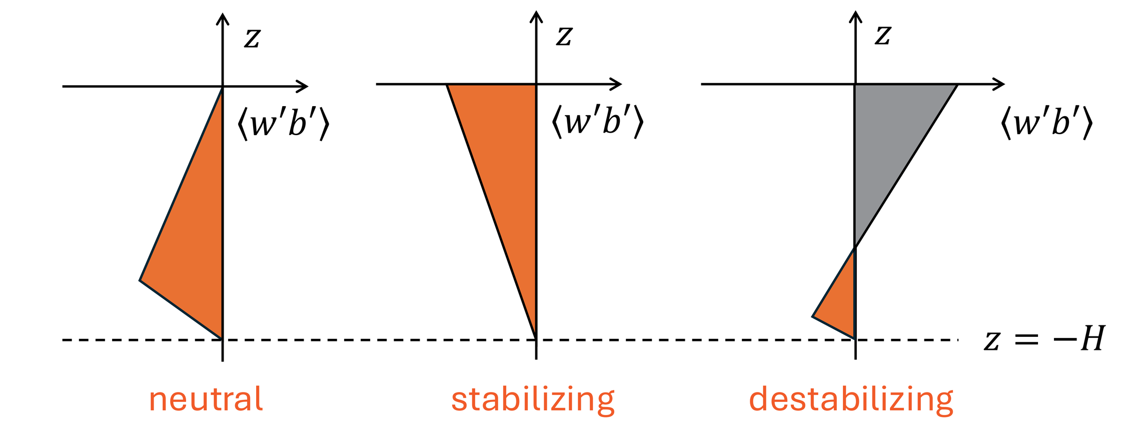
1. Introduction

We aim to find **a new formulation** for the energetic Planetary Boundary Layer (ePBL) that can simulate **the transient response** of the boundary layer (H) to surface forcing. A neural network (**FCNN_trans**) is trained to predict the time-dependent rate of potential energy conversion (M) in a series of single column General Ocean Turbulence Model (GOTM) simulations driven by constant surface forcing. The emulator's performance is compared to a predictor trained on equilibrium M values, **FCNN_mean**.

2. ePBL

Figure 1: Potential energy conversion to turbulent kinetic energy under neutral, stabilizing and destabilizing surface buoyancy flux. Adapted from [1].



$$-M = \int_{-H}^{0} \min(\langle w'b' \rangle, 0)\, dz$$

$$M = \underbrace{(m_* + m_{*LT})u_*}_{\text{mechanical processes}} + \underbrace{n_* \int_{-H}^{0} \max(\langle w'b' \rangle, 0)\, dz}_{\text{convective processes}}$$

→ H is predicted from a parameterization for **the rate of potential energy production per unit area, M**.

→ $n_* = 0.066$; m_* is a function of the surface forcing and the boundary layer depth, and m_{*LT} is a function of the Langmuir number.

→ **These functional forms are determined using $k - \varepsilon$ simulations assuming fully equilibirated boundary layer.**

3. Data

Model used: GOTM [2] $k - \varepsilon$ parameterized as in [3].
Duration and temporal resolution: 9 days, 30 minutes.

FCNN_mean is trained on **$\overline{M}$ and $\overline{H}$ averaged** over the last day of their respective timeseries. **FCNN_trans** is trained on **all points in the timeseries**, exculding the first inertial period.

TEST DATA FORCING

216 MEAN DATAPOINTS
82 674 TRANSIENT DATAPOINTS

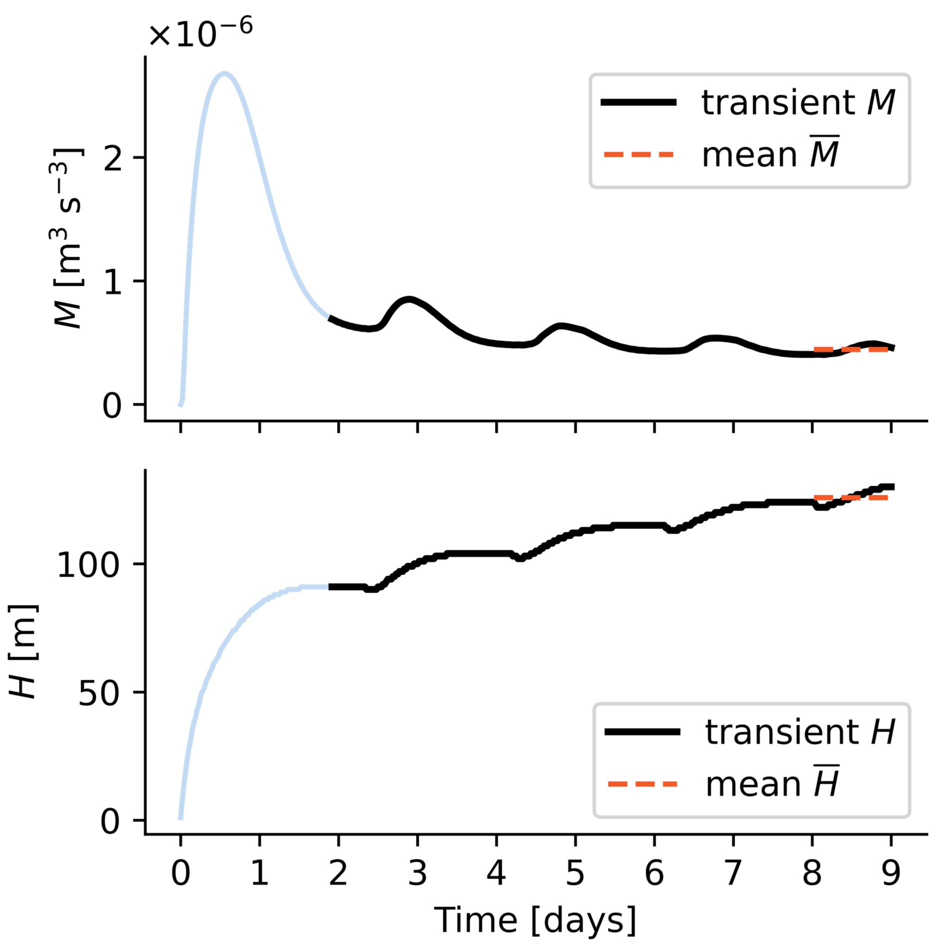
Initial $\partial\theta/\partial z$ [°C m^{-2}]	0.005, 0.015, 0.03
Latitude [°N]	15, 25, 55, 85
Heat flux [W m^{-2}]	-90, -40, -10, 10, 40, 90
Wind stress [m^2 s^{-2}]	0.25, 0.75, 0.95

TRAINING DATA FORCING

3 240 MEAN DATAPOINTS
1 250 280 TRANSIENT DATAPOINTS

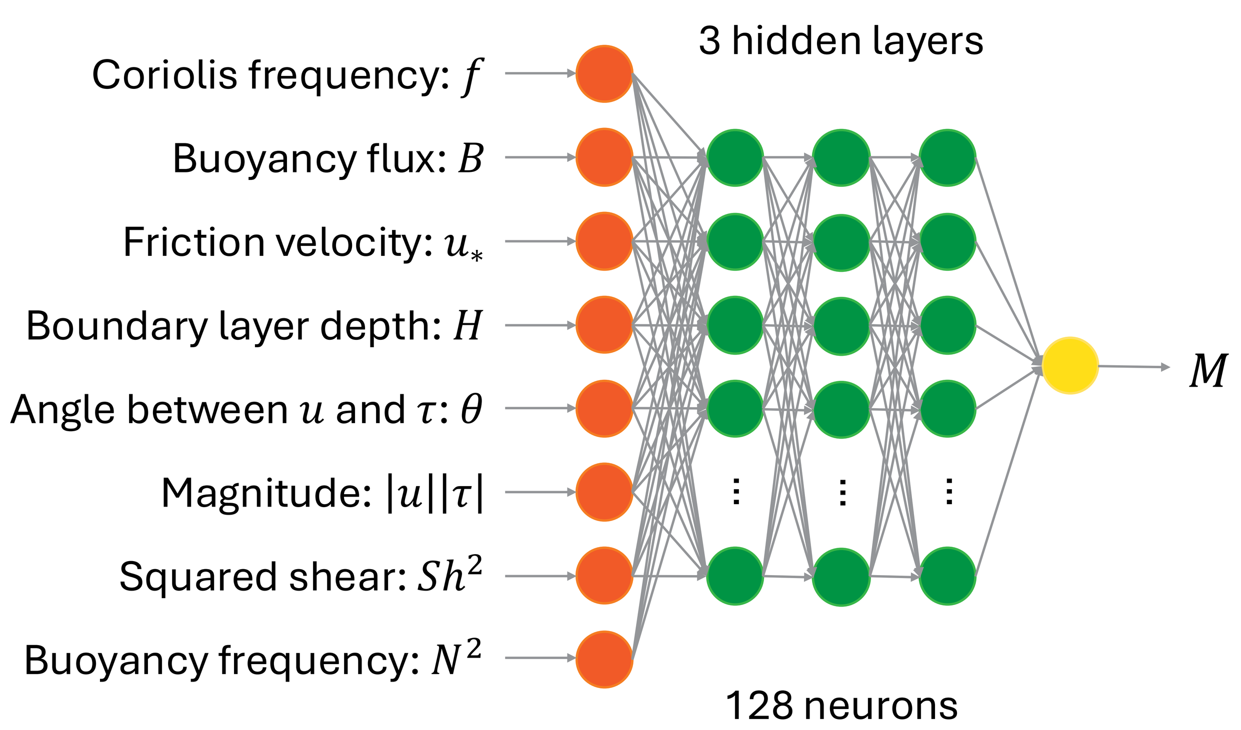
Initial $\partial\theta/\partial z$ [°C m^{-2}]	0.001, 0.01, 0.02, 0.04
Latitude [°N]	10, 20, 30, 40, 50, 60, 70, 80, 90
Heat flux [W m^{-2}]	-100, -75, -50, -25, 0, 25, 50, 75, 100
Wind stress [m^2 s^{-2}]	0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0

Figure 2: An example timeseries of M and H for constant f, B and u_*.

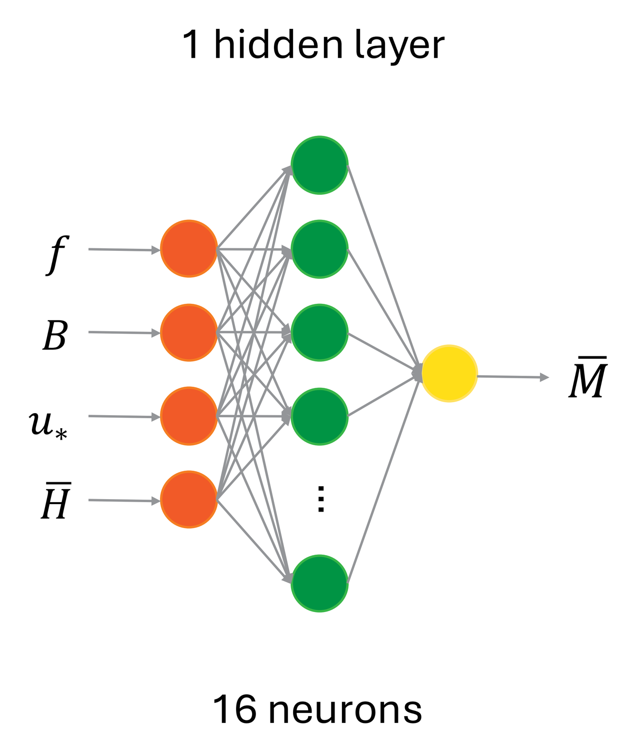


4. Results

FCNN_trans



Benchmark: FCNN_mean



Why these inputs? The squared shear and buoyancy frequency are sampled at $z = H$.

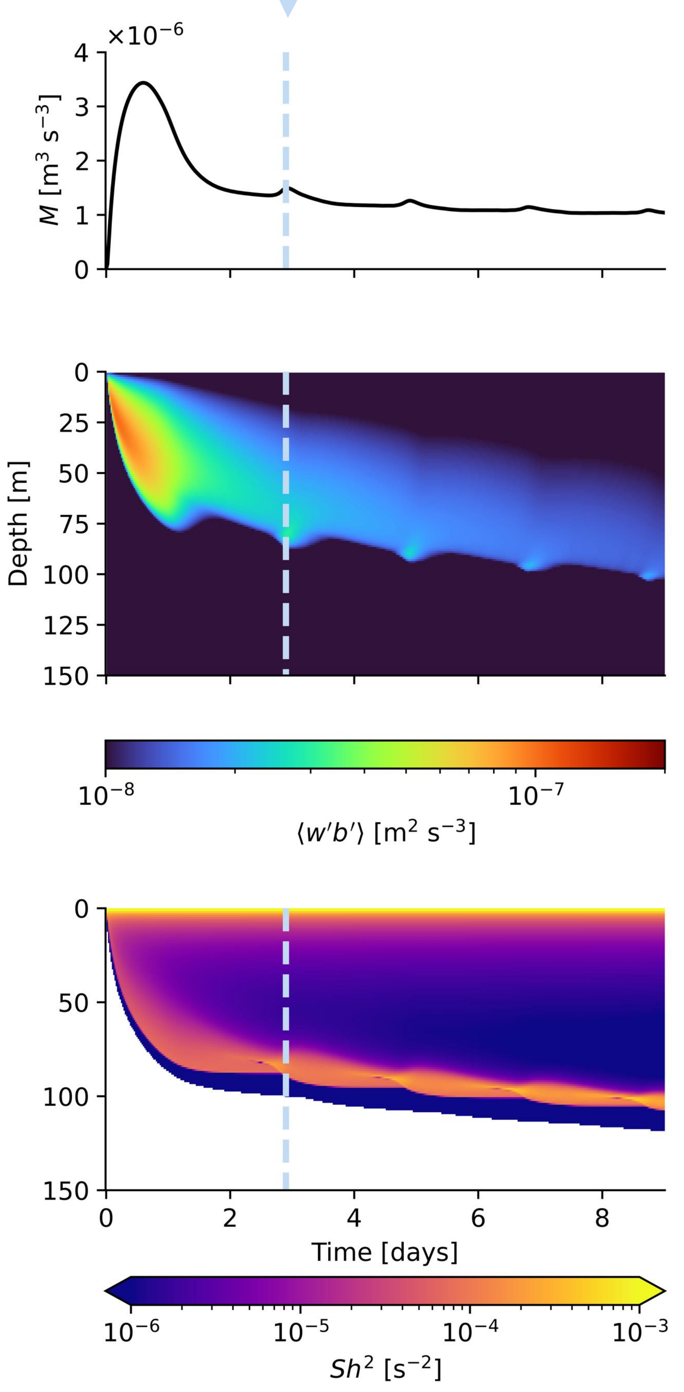


Figure 3: An example of timeseries of M, turbulent buoyancy flux and squared shear. The vertical dashed line shows that the inertial oscillations of M correspond to elevated shear at the boundary layer base.

Two Fully Connected Neural Networks are trained to predict M. The benchmark, FCNN_mean, **mimics the current algebraic ePBL parameterization.** The extra inputs to FCNN_trans enable the network to reproduce the inertial oscillations seen in M timeseries (Fig. 3 and 4). **The transient M is predicted 68% more accurately by FCNN_trans compared to FCNN_mean (Fig. 5).**

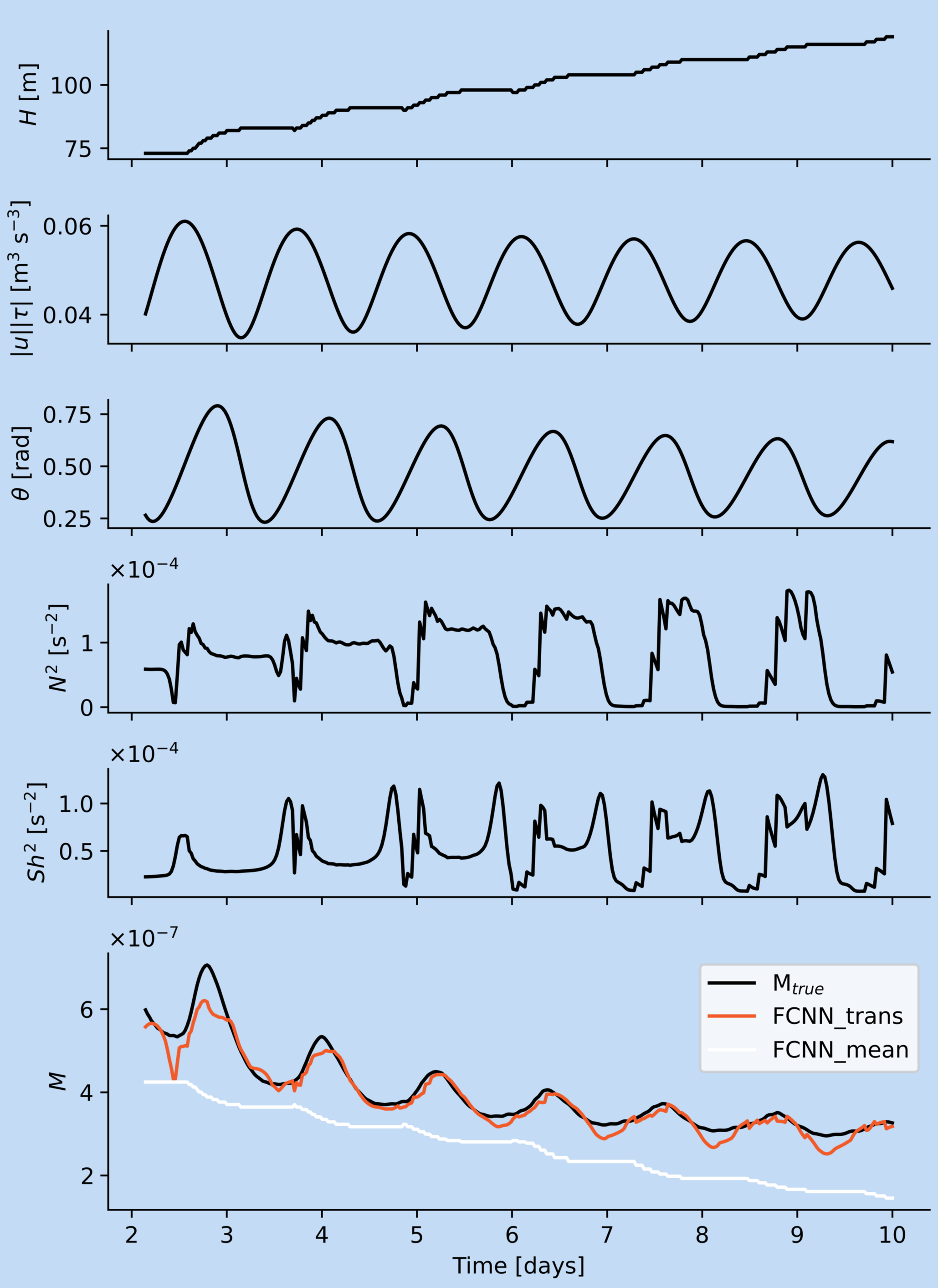
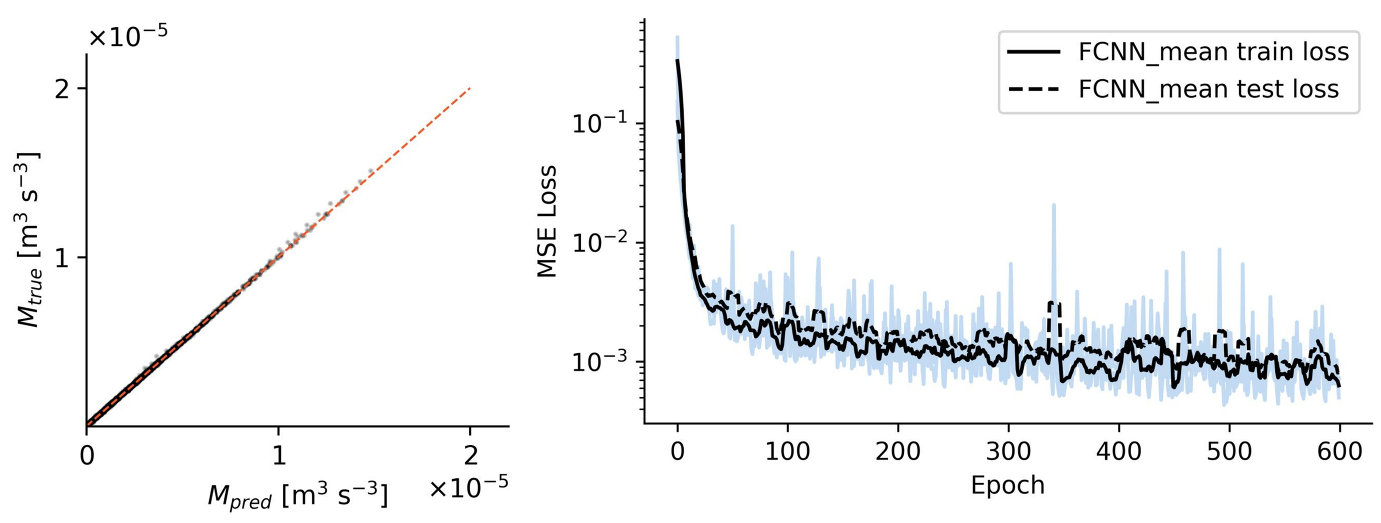
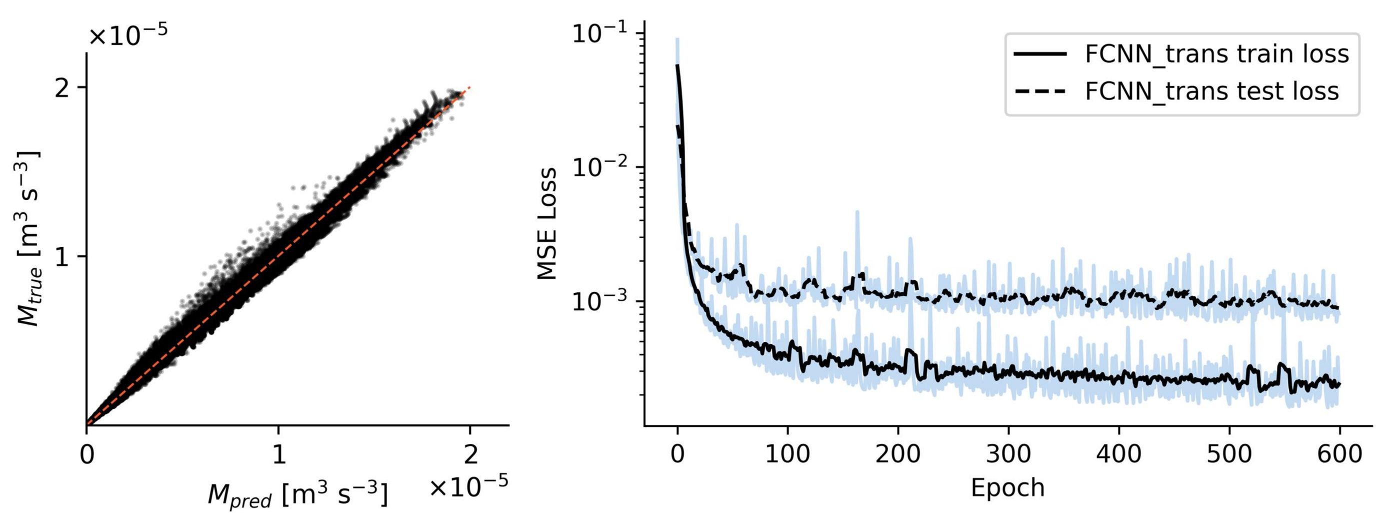


Figure 4: An example of a comparison between M prediction by FCNN_mean and FCNN_trans.

FCNN_mean PREDICTION OF $\overline{M}$



FCNN_trans PREDICTION OF M



FCNN_mean PREDICTION OF M

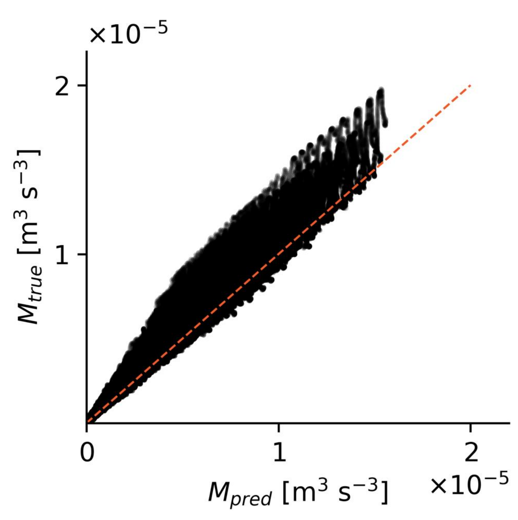


Figure 5: The training and prediction skills of FCNN_mean and FCNN_trans. The RMSE of FCNN_trans prediction of M is $2.1 \cdot 10^{-7}$ **m³ s⁻³; the RMSE of FCNN_mean prediction of M is** $6.5 \cdot 10^{-7}$ **m³ s⁻³.**

5. Next steps

In this work, it is assumed that the training data generated with the $k - \varepsilon$ model can be used as a proxy for Large Eddy Simulations (LES). **This assumption breaks in the presence of Langmuir turbulence. The surface wave impact on M has to be learned from LES or using an alternative SMC parameterization.**

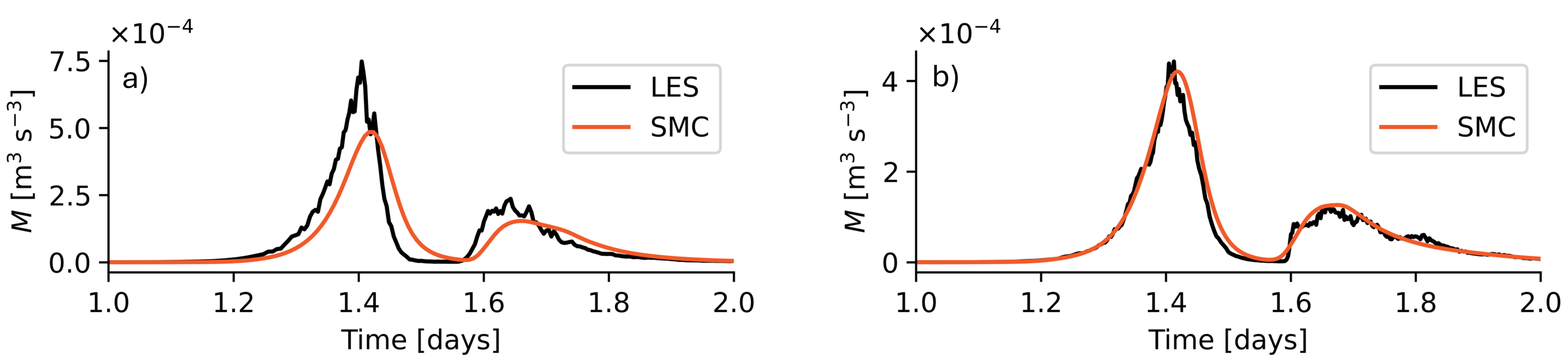


Figure 6: M response to an idealized hurricane in LES and an SMC model a) with and b) without surface waves.

REFERENCES: [1] Reichl and Li (2019), AMS. [2] Umlauf, Burchard and Bolding (2005), gotm.net [3] Reichl and Hallberg (2018), Ocean Modelling.